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Prof. Kenneth M. Merz, Jr., Editor-in-Chief

Journal of Chemical Information and Modeling

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Dear Prof. Merz,

We submit our revised manuscript “**Computational Discovery of Antimalarial Candidates from African Natural Product Chemical Space**” for consideration as an Article in *JCIM* (Manuscript ID: ci-2026-012015).

This work presents a computationally benchmarked generative scaffold-hopping framework for antimalarial discovery. Rather than a static hit list, the framework retrospectively predicts (retrodicts) known ground-truth antimalarials from the MMV Malaria Box before application to African chemical space, achieving 69.8% hit rates and consensus-score ROC-AUC 0.924–1.000. Among VAE-generated molecules, 92.6% access chemotypes unreachable by local SELFIES perturbations of known drugs (mean max Tanimoto 0.206), and the $1.84\times$ scaffold-to-whole-molecule Tanimoto ratio indicates systematic ring-system preservation.

The three-component pipeline—scaffold-aware VAE sampling, dual-filter consensus docking (DiffDock + AutoDock Vina), and multi-parameter optimization—reduces screening costs by 99.3%. All validation protocols are documented with open-source code, and the full dataset is deposited on Zenodo (10.5281/zenodo.19608875).

Suggested reviewers.

1. Dr. Maria Sorokina, University of Jena — natural product cheminformatics
2. Dr. Ntie-Kang Fidele, University of Buea — African NP virtual screening
3. Dr. François Dèret, Medicines for Malaria Venture — antimalarial screening

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke Vital Sao Temgoua

(on behalf of all co-authors)